

SYLLABUS FOR PRECLINICAL STUDIES

MODULE I

BODY STRUCTURE, ORGANIZATION AND FUNCTIONS-II (BSOF 1012)

MODIFIED MARCH, 2022

MODULE INFORMATION

I. Module Title: Body Structure, Organization and Function-II

II. Module Code: BSOF 1012

III. Credit Value (ECTS): 6

IV. Duration: Year I, Semester I

V. Prerequisites: Premedical courses

VI. Module Description

This module is designed for first year undergraduate medical students and is intended to describe body structure and function, as well as molecular mechanisms that operate in the living cell. It also deals with developmental periods, with an emphasis on the beginning of human development, formation of embryonic discs and organogenesis, the fetal period, human birth defects, formation of all body systems and molecular regulation of development.

VII. Module Contents and Learning Outcomes

Module objectives:

At the end of the module the preclinical students will be able to:

- Recognize the body's structure, organization and function, including molecular mechanisms that operate in the living cell
- Dissect cadavers, use and operate light microscopes, learn basic histological techniques and perform biochemical tests in the living system
- Recognize that physiology is a science involving adjustment mechanisms to prevailing environmental conditions and understand the concepts of homeostasis, allostasis and heterostasis

Supportive objectives of the Module

At the end of the module the preclinical students will be able to:

- Interpret the potential spaces in the body cavities and the arrangement of their membranes
- Describe the structure and functions of skin, superficial fascia and subcutaneous tissue
- Compare the processes of bone formation and resorption
- Explain the components of the skeletal muscles of the back and upper limb
- Follow the guidelines for dissection of the back and upper limb regions
- Trace the proximal and distal attachments, innervations and vascularization of each muscle in the back and upper limb too

- Trace the vertebrae, curvatures, ligaments and joints of the backbone
- Describe the sites of safe lumbar puncture
- Describe common bone fractures and their management
- Identify the superficial, intermediate and deep back muscles, along with their blood and nerve supply
- Identify the anterior and posterior compartment muscles of the upper limb, along with their blood and nerve supply
- Observe how tissue sections are prepared for light microscopic examination
- Describe the organization of cells and extracellular matrix in the tissues, organs and organ systems
- Demonstrate knowledge of the sequence of development from fertilization, formation of tissue organs and organ systems
- Identify developmental malformations of organs and organ systems.
- Demonstrate understanding of the interaction of cells for induction, determination and differentiation in the formation of organs and organ systems, as well as molecular regulation for such interactions
- Describe genetic and environmental factors that govern the development of the human body
- Describe the basic principles of physiological adjustment.
- Describe the distribution of fluid and electrolyte across cell membranes
- Describe transport mechanisms across cell membranes
- Describe the autonomic nervous system
- Understand the bioelectricity of cells
- Explain autonomic control of visceral function

VIII. Learning and Teaching Strategies

8.1. The strategies include:

- Small group learning
- Practical learning
- Demonstrations

8.2. Methods of Teaching and Learning:

- Standard Lecture
- Seminars
- Tutorials
- Demonstration
- Practical learning
- Dissections

8.3. Human resources:

- **Staff:** Instructors and graduate assistants drawn from the department of anatomy, and physiology will be involved in lecture delivery, laboratory demonstration, facilitation and assessment of seminars, tutorials, quizzes and tests.
- **Spaces/location:** lecture halls (Sefere Selam), labs, dissection rooms (Black Lion)

- **Instructional media:** interactive white board, printed materials (handouts, books, and work sheets), blackboards, display board, flip charts, slides, overheads, real objects or models, images, projector, LCD projectors, DVDs and CD-ROMs will be used during lectures, seminars and tutorials. The laboratory session requires ball and stick models of different atoms, pH meters and solutions of various pH strengths for biochemistry; forceps, scissors, blades, wood block, dissection table etc for anatomy; microscope, images for histology; preserved embryonic specimens for embryology.
- **Time:** Lecture: Monday to Friday from 8:30-12:10 a.m.
Laboratory: Monday to Friday 1:30 - 04:30 p.m.

IX. Role of Instructors:

The role of instructors includes giving standard lectures, seminars and assignment; involving learners fully in assessments, help students to understand what is expected, addressing students questions, facilitating discussion, and developing learners roles in moderation activities. It is also to instructors' role to plan, design and carryout the lesson plan, to collaborate with physiology staff to accomplish the module successfully, to dissect cadavers, demonstrate prosected cadaver anatomy, to evaluate students and to correct quizzes, tests and exams.

X. Role of learners:

It is the learners' role to engage actively in learning, attend lectures, participate actively, share ideas with each other, ask questions on unclear issues, do seminars and assessments, choose topics for tutorials, refer to different reading materials, review their own learning through self-assessment and to demonstrate their knowledge, skill, attitudes and capabilities through a range of evidence including self-assessment, quiz, tests, activities, seminars and examination.

XI. Assessment methods

Assessment criteria of anatomy, embryology, histology, and physiology

- Two tests – 50%
- Final examination- 50%

Body Structure, Organization and Function-II

Module Name	Body Structure, Organization and Functions-II
Module number	BSOF 1012
ECTS	6
Coordinator	Yossef Teshome (Department of Anatomy) Email: yossef.teshome@gmail.com Phone: +251-944126949
Collaborator	Mr. Tekalgn (Department of Physiology) Phone: +251-912020304
Course Schedule	January 16 – February 24, 2023
Duration of the course	6 weeks including exam week
Number of exams	Two progressive tests (50 questions each) Final exam (100 questions)
Placement for lectures	Sefere Selam campus
Practical placements	DR (Anatomy department Room-I and Room-II) Histology lab (Anatomy department Room-I)

COURSE CALENDAR FOR MODULE-I: BSOF-II

MODULE I: BSOF II (Week 1-6)

MODULE ONE: BSOF II THEME ONE TIME TABLE FOR WEEK ONE (January 16 – January 20/2023)

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
8:30:-10:10 AM INSTRUCTOR	Anatomy: Back anatomy: General description; key features; components. Mr Yossef	Anatomy: Back musculature, and Neurovasculature bundles Mr Yossef	Anatomy: Introduction to the upper limb, Bones of the upper limb and Joints Mr Fikre	Epiphany	Anatomy: Pectoral region & the Shoulder region, Mr Fikre
10:10-10:30 AM	BREAK				
10:30:-12:10 AM INSTRUCTOR	Anatomy: Typical and atypical vertebrae. Back ligaments and joints Mr Yossef	Anatomy: Clinical correlation; Lumbar puncture; disc prolapse Mr Yossef	Anatomy: Superficial structures of upper limb & Breast Mr Fikre	Epiphany	Anatomy: Axilla, Boundaries, Contents, communications Mr Fikre
12:00-1:30 PM	LUNCH				
1:30-4:30 PM INSTRUCTOR	DR1: Group 1 DR2: Group 2 Dr. Solomon & Mr. Fikre	DR1: Group 3 DR2: Group 4 Dr Soressa & Mr. Yossef	DR1: Group 5 DR2: Group 6 Mrs. Shewit & Mr. Abay	Epiphany	
INSTRUCTOR					

MODULLE ONE
BSOF II
TIME TABLE FOR WEEK TWO (January 23 – January 27/2023)

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
8:30:-10:10 AM INSTRUCTOR	Histology: Epithelial Tissue1: Form & Characteristics Mr. Reta	Anatomy: Axilla, Boundaries, Contents, communications Mr Fikre	Anatomy: Forearm (Anterior & Posterior compartments) Mr Fikre	Histology: Connective tissue proper 1 Mr. Reta	Histology: Connective tissue 2: Special CT Mr. Reta
10:10-10:30 AM	BREAK				
10:30:-12:10 AM INSTRUCTOR	Histology: Covering & Glandular Epithelium Mr. Reta	Anatomy: Arm (Anterior & Posterior compartments& Cubital fossa Mr Fikre	Anatomy: Hand (dorsum and palm of the hand) Mr Fikre	Histology: Connective tissue proper 1: Mr. Reta	Histology: Connective tissue 2: Special CT Mr. Reta
12:10-1:30 PM	LUNCH				
1:30-4:30 PM INSTRUCTOR	DR1: Group 1 DR2: Group 2 Mr. Fikre & Mr. Abay	DR1: Group 3 DR2: Group 4 Mr. Yossef & Mrs. Shewit	DR1: Group 5 DR2: Group 6 Dr. Solomon & Mr. Yossef	DR1: Group 1 DR2: Group 2 Mrs. Shewit & Mr. Fikre	DR1: Group 3 DR2: Group 4 Mr. Abay & Dr. Soressa
INSTRUCTOR	Hist. Lab 1: Group 3 1:30- 3:30 PM Hist. Lab 2: Group 4 3:30- 5:30 PM Dr. Soressa	Hist. Lab 1: Group 5 1:30- 3:30 PM Hist. Lab 2: Group 6 3:30- 5:30 PM Dr. Solomon	Hist. Lab 1: Group 1 1:30- 3:30 PM Hist. Lab 2: Group 2 3:30- 5:30 PM Mrs. Shewit	Hist. Lab 1: Group 3 1:30- 3:30 PM Hist. Lab 2: Group 4 3:30- 5:30PM. Prof. Mekbeb	Hist. Lab 1: Group 5 1:30- 3:30 PM Hist. Lab 2: Group 6 3:30- 5:30 PM Prof. Mekbeb
1:30-4:30 PM Clinical attachment			Group 3 & 4 Ortho	Group 5 & 6 Ortho	Group 1 & 2 Ortho

MODULLE ONE: BSOF II
THEME TWO
TIME TABLE FOR WEEK FOUR (February 6 –February 10 /2023)

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
8:30:-10:10 AM INSTRUCTOR	TEST I (50) Anatomy (30) & Histology (20) Invigilators (Anatomy staff)	Embryology: Human Birth Defects I Human Birth Defects II Mr. Worku	Physiology: Excitable Tissue Mr. Tekalgn	Physiology: Excitable Tissue Mr. Tekalgn	Physiology: Autonomic Nervous System: Introduction; Physiologic anatomy; Mr. Tekalgn
10:10-10:30 AM	BREAK				
10:30:-12:10 AM INSTRUCTOR		Embryology: Development of Body Cavities, Mesenteries, and Diaphragm, Malformations Mr. Worku	Physiology: Excitable Tissue Mr. Tekalgn	Physiology: Excitable Tissue Mr. Tekalgn	Physiology: ANS Receptors of effector organs Activity pattern Mr. Tekalgn
12:10-1:30 PM	LUNCH				
1:30-4:30 PM		DR1: Group 1 DR2: Group 2 Mr. Yossef & Mrs. Shewit	DR1: Group 5 DR2: Group 6 Dr. Solomon & Mr. Fikre		
		Hist. Lab 1: Group 5 Hist. Lab 2: Group 6 Mr. Fikre	Hist. Lab 1: Group 3 Hist. Lab 2: Group 4 Dr. Soressa	Hist. Lab 1: Group 1 Hist. Lab 2: Group 2 Mr. Abay	

MODULLE: BSOF II THEME TWO TIME TABLE FOR WEEK FIVE (February 13– February 17/2023)					
TIME	MONDAY	THUESDAY	WEDNES DAY	THURSDAY	FRIDAY
8:30:-10:10 AM Instructor	Physiology: Autonomic functions after spinal cord transaction Mr. Tekalgn	Study Period			TEST II (50) Embryology (28) and Physiology (22) Invigilators (Anatomy & Physiology staff
12:10-1:30 PM		LUNCH			
1:30-4:30 PM Instructor	Physiology: Autonomic functions after spinal cord transaction Mr. Tekalgn	Study Period			

MODULLE: BSOF II

THEME TWO TIME TABLE FOR WEEK SIX (February 20-February 24/2023)					
TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
8:30:-10:10 AM Invigilators	Study Period				FINAL EXAM (100) Anatomy (32) Histology (21) Embryology (26) Physiology (21) Anatomy & Physiology staff
10:10-10:30 AM					
10:30:-12:10 AM					
1:30-4:30 PM					

VIII. Reading Resources

1. Moore KL, Dalley AF. Clinically Oriented Anatomy, 8th ed. Baltimore: Lippincott Williams & Wilkins, 2013
2. Henry Gray and Henry Vandyke Carter: Grey's Anatomy, 40th ed. Church. Livingston, London. 2008
3. Robert H. Whitaker and Neil R. Borley: Instant Anatomy, 4th ed. 2010
4. Frank H Netter: Atlas of Human Anatomy, 6th ed. Saunders, 2014.
5. Ross MH, Romrell, LJ & Kaye, G.I. Histology. A Text and Atlas. 6th ed. Williams & Wilkins. London
6. Junqueira, L.C., Carneiro, J. & Kelley, R.O. Basic Histology. 14th ed. Appleton & Lange. 2015
7. DiFiore's Atlas of Histology with Functional Correlations. Williams & Wilkins. London
8. Moore, K.L. The Developing Human: Clinically Oriented Embryology, 9th ed. W. B. Saunders, Philadelphia. 2011
9. Sadler, J.W. Langman's Medical Embryology, Williams & Wilkins, Baltimore.
10. Dissection Manual: Romanes, G.J., Cunningham's. with different volumes available in the Department of Anatomy
11. Textbook of Medical physiology, A. Guyton, 10th edition
12. Review of Medical physiology, William F. Ganong, 16th edition
13. Human Physiology, Seven Edition. Vander Sherman Luciano
14. Weaver, Robert F. *Molecular Biology*, 2nd ed. New York : McGraw-Hill, 2002.

IX. Lecture Topics

Theme	LECTURE TOPICS
	Anatomy
I	THEME ONE: Back (8 hrs) 1.1. General description; key features; components: skeletal, spinal nerves, surface anatomy; curvatures. 1.2. Typical and atypical vertebrae 1.3. Back ligaments and joints 1.4. Back musculature 1.5. Clinical correlation; Lumbar puncture; disc prolapse (to be taught together with clinicians)
I	THEME ONE: Upper Limb (16hrs) 2.1. General introduction to the upper limb. 2.2. The bones of the upper limb 2.3. Surface anatomy of the free upper limb – subcutaneous vessels and nerves 2.4. Pectoral region and breast 2.5. Axilla boundaries and contents 2.6. Posterior compartment of the arm and forearm and dorsum of the hand 2.7. Anterior compartment of the arm and forearm; the cubital fossa

	2.8. Palm of the hand 2.9. The joints of the pectoral girdle, elbow, wrist joints and other joints of the forearm and hand
I	THEME ONE: Histology (16 hrs) 3.1. Epithelial Tissue 1: Forms & Characteristics 3.2. Epithelial tissue 2: Covering & Glandular Epithelium 3.3. Connective tissue 1: Connective tissue proper 3.4. Connective tissue 2: Special Connective Tissue 3.5. Nervous Tissue 1: Neurons 3.6. Nervous tissue 2: Neuroglia 3.7. Peripheral Nervous System
I	THEME ONE: Embryology (20 hrs) 4.1. First Week of Development, Second Week of Development 4.2. Third Week of Development 4.3. Embryonic Period 4.4. Fetal Period 4.5. Extra-embryonic Membranes and Placenta 4.6. Multiple Pregnancy 4.7. Human Birth Defects I 4.8. Human Birth Defects I 4.9. Development of Body Cavities, Primitive Mesenteries, and the Diaphragm, Malformations
II	THEME TWO: Physiology (16 hrs)
	5. Cell excitation – 8 hours

	6. Autonomic Nervous System -4 hours 6.1. Introduction to ANS Physiologic anatomy of ANS Basic characteristics of the SANS and PANS Cholinergic and adrenergic fibers Mechanisms of transmitter secretion and removal
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	Receptors of the effector organs Acetylcholine (muscarinic, nicotinic) Adrenergic receptors (alpha and beta) 6.2. The presynaptic control of transmitter release Effects on specific organs Function of the adrenal medulla Activity pattern of ANS Sympathetic and parasympathetic tone Effects of autonomic denervation Denervation supersensitivity Autonomic functions after spinal cord transaction
	7. Autonomic functions after spinal cord transaction – 4 hrs